TRENTHAM TRIBOLOGY LABORATORY
Independent Materials & Tribology Testing | Trentham Science Park, Stoke-on-Trent ST4 8GH

EXPERT REPORT OF DR HELEN TRENTHAM CEng FIMMM

In the matter of Harrow Precision Engineering Limited v Melbourne Sealtech Limited — Claim No. HT-2025-000318

Report dated 12 September 2025 | Ref: TTL-25-0417

1. Introduction and instructions

1.1 I am Dr Helen Trentham, principal consultant at Trentham Tribology Laboratory. I hold a PhD in tribology and materials science and am a Chartered Engineer and Fellow of the Institute of Materials, Minerals and Mining. I have 22 years' experience in the failure analysis of mechanical seals and rotating equipment.

1.2 I am instructed by Whitfield Carter LLP on behalf of the Claimant to examine failed and unused Type MS-72 cartridge seals and to determine (a) whether the seal-face material conforms to grade CG-218; (b) whether the lapped face flatness conforms to the specified two-helium-light-band tolerance; and (c) the cause of the premature in-service failures.

1.3 I understand my overriding duty is to the Court and not to those instructing me. I have complied with that duty and with CPR Part 35 and the associated Practice Direction and Guidance.

2. Materials examined

2.1 I examined 18 failed seals recovered from HP-400 pumps in the field and 12 unused seals drawn from the Claimant's stock, all identified as Melbourne Sealtech Type MS-72. I also reviewed drawing HP-400/SEAL Rev C, the Claimant's internal investigation QIR-2024-118, and the field failure reports.

3. Methodology

3.1 I carried out: (a) visual and stereo-microscopic examination of the seal faces; (b) monochromatic (helium) light-band interferometry to measure face flatness; (c) glow-discharge optical emission spectroscopy and image analysis of the carbon-graphite to determine antimony impregnation content and porosity; and (d) scanning electron microscopy of the failure surfaces.

4. Findings — material grade

4.1 Grade CG-218 requires an antimony impregnation of 10–14% by volume of the accessible porosity. The failed and unused MS-72 faces showed antimony impregnation in the range 3.2–5.1%, materially below specification. The carbon matrix also showed higher residual open porosity than grade CG-218 permits.

4.2 The low impregnation reduces the thermal conductivity and mechanical strength of the face and predisposes it to heat-checking (thermal cracking) and accelerated wear under normal running conditions.

5. Findings — flatness

5.1 The specification requires lapped face flatness within two helium light bands (approximately 0.58 µm). Of the 12 unused seals measured, 9 exceeded two light bands and 5 exceeded six light bands

(approximately 1.7 µm). Faces this far out of flat cannot maintain the hydrodynamic fluid film required for reliable sealing.

6. Findings — failure surfaces

6.1 The failed faces exhibited radial heat-check cracking, glazing and localised pull-out of the carbon, consistent with dry or marginal running caused by loss of the fluid film. I found no evidence characteristic of abrasive third-body wear that would indicate operation on heavily contaminated water; the wear pattern is that of thermal/adhesive distress originating at the face, not ingress of abrasive particles.

7. Opinion on causation

7.1 In my opinion, on the balance of probabilities, the premature failures of the MS-72 seals were caused by the combination of (a) under-specified antimony impregnation of the carbon-graphite face material and (b) out-of-tolerance face flatness, both of which are attributes of the seals as manufactured and supplied, and not of their installation or application.

7.2 The consistency of the material and flatness deficiencies across both failed and unused samples indicates a systemic manufacturing deficiency rather than isolated or application-induced failures. The alternative causes of misapplication and mis-installation advanced by the Defendant are not, in my opinion, supported by the physical evidence.

8. Declaration

8.1 I confirm that I have made clear which facts and matters referred to in this report are within my own knowledge and which are not. I confirm that the opinions I have expressed represent my true and complete professional opinion. I understand my duty to the Court and have complied with it.

Signed: *H. Trentham*

Dr Helen Trentham CEng FIMMM — Trentham Tribology Laboratory — 12 September 2025